Research Statement

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I am a macroeconomist working at the intersection of finance, firm dynamics, and monetary economics. My research agenda is organized around a central question: how does financial intermediation shape the allocation of resources and, through that allocation, economic activity and policy? I approach this question with two related strands of work. The first studies banks as heterogeneous firms and examines how their dynamics shape the allocation of credit across borrowers, economic activity, and monetary policy transmission. The second studies how financial conditions shape the extensive margin of firm dynamics: who becomes an entrepreneur, where firms enter, and how new firms subsequently grow. Throughout my research, I use structural models to address measurement problems and guide empirical estimation from microeconomic data to aggregate outcomes. I then use these estimates to discipline quantitative models of economic mechanisms and policies.

Bank dynamics, real activity, and policy

Bank lending is central to both short-run fluctuations and long-run growth because of its role in determining which firms receive credit and on what terms. Banks, however, are themselves heterogeneous firms whose lending decisions vary with their productivity, funding costs, and the demand, whereas canonical macroeconomic models typically represent intermediation through a representative bank. Consequently, aggregate outcomes depend not only on how much credit is supplied, but also on which banks supply it and, in turn, which firms receive it. My research combines the macroeconomics of credit with the study of firm dynamics, modeling banks as firms and studying how their dynamics shape the allocation of credit and economic activity.

In ***"Banks as Firms: The Macroeconomics of Financial Firm Dynamics,"*** my job market paper, I develop a framework for bank dynamics that combines insights from the non-financial firm dynamics literature with a model of banking competition. In this setting, banks grow by expanding across regional credit markets and capturing market share based on differences in their ability to screen risk, attract borrowers, and source capital. Guided by this theory, I estimate these bank-level fundamentals using a structurally identified dynamic state-space model. This measurement exercise yields bank-level estimates of lending productivity and bank-specific demand, objects that are central to the study of non-financial firms but have not previously been measured for banks. I then embed these estimates in a quantitative multi-region New Keynesian model to quantify how heterogeneity in bank fundamentals shapes bank dynamics, credit allocation, and aggregate outcomes.

I find that banks' screening ability, a measure of lending productivity, is the primary driver of bank dynamics, accounting for three-quarters of a bank's idiosyncratic loan growth. Screening ability is particularly important because it affects both lending decisions and loan losses, so deterioration in screening can trigger an accelerating balance-sheet contraction and amplify bank-level shocks.

Bank dynamics then influence the economy through two channels: the allocation of credit across firms and the transmission of monetary policy. First, the distribution of screening ability affects how effectively the banking sector routes resources to productive firms. When screening

deteriorates, banks direct more credit toward firms that exit before production, reducing aggregate labor productivity. Second, because banks differ in their fundamentals and operate across different regional credit markets, monetary policy also has heterogeneous regional effects. Regions more exposed to banks with weaker fundamentals experience larger contractions in lending following a monetary tightening, linking the composition of local lenders to both aggregate monetary transmission and the regional distribution of economic activity.

In complementary work, ***"Banks, Sentiments, and Business Cycles"*** (under review), I study another source of heterogeneity in financial intermediation: banks' beliefs about future credit risk. To do so, I use insights on measuring productivity from the firm dynamics literature to develop an estimation procedure for measuring bank sentiments —a bank's irrational assessment of the future default rate in its loan portfolio. I then incorporate bank sentiment into an otherwise standard Bayesian vector autoregression model and trace the effects of sentiment shocks on the broader economy. I find that pessimistic bank sentiment shocks act like a negative credit supply shock propagating primarily through the business lending channel, raising borrowing costs and tightening lending standards independent of realized risk. Moreover, monetary policy can only partially offset these effects. Together with my job market paper, this work shows how both bank fundamentals and bank beliefs can affect the supply and allocation of credit.

In earlier work, ***"Bank Lending Standards and the U.S. Economy"*** (with Elijah Broadbent, Huberto Ennis, and Horacio Sapriza; published in the *Journal of Economic Dynamics and Control*), we study how changes in bank lending standards affect aggregate activity. We identify bank lending standard shocks from confidential bank-level responses to the Federal Reserve's Senior Loan Officer Opinion Survey and embed these shocks in a macroeconometric vector autoregression. We find that contractions in lending standards significantly slow GDP growth and induce a monetary policy response, with the effects concentrated in corporate lending. However, the magnitude of these effects varies with economic conditions: lending-standard shocks have larger effects when output is below trend, the policy rate is away from its lower bound, or economic uncertainty is elevated. This work provides an earlier foundation for my subsequent research by establishing the macroeconomic importance of bank-level lending conditions and motivating the study of the underlying heterogeneity and dynamics of banks themselves.

<u>The real effects of entrepreneurial finance</u>

A second strand of my research studies how financial conditions shape the extensive margin of firm dynamics: the emergence, location, and subsequent growth of new firms. This work complements my research on bank dynamics by examining how financial conditions affect not only the financing of existing firms, but also how firms enter the economy in the first place.

In ***"Uncertainty, Market Concentration, and the Entrepreneurial Funding Channel,"*** I document that uncertainty shocks about non-financial productivity have persistent effects on economic activity that standard uncertainty transmission mechanisms cannot account for. I develop a macroeconomic model with a new entrepreneurial funding channel in which risk-averse financial institutions become more reluctant to finance prospective entrepreneurs following an increase in uncertainty. Firm entry therefore falls, creating a missing cohort of firms. When the economy recovers, entry does not fully rebound, so the missing cohort persists and reduces the number of firms in the economy. The resulting decline in firm entry dampens output persistently and increases

market concentration among incumbent firms. The implication is that uncertainty can affect the long-run composition of the firm population and generate persistent changes in market structure.

In related new work, ***"Local Financial Shocks and Business Formation: Entry, Startup Scale, and Growth"*** (with Emin Dinlersoz, Timothy Dunne, John Haltiwanger, and Veronika Penciakova), we use U.S. Census Bureau and IRS administrative data to study how financial conditions shape where entrepreneurship occurs and how new firms perform. We separately identify the effects of household wealth shocks and bank credit supply shocks on nascent entrepreneurship and post-entry firm growth. The goal is to understand how financial conditions at the point when an entrepreneur develops a business idea affect both the spatial distribution of startups and their subsequent lifecycle dynamics. By distinguishing the effects of household balance sheets from local credit supply, the project provides evidence on the mechanisms through which financial conditions shape the composition and geography of new firms.

Future research

My future research will study how heterogeneity in financial intermediation shapes the allocation of capital across firms and regions, and how that allocation interacts with monetary and financial policy. My job market paper provides a starting point: a framework in which banks endogenously expand across and compete in geographic credit markets.

One project will study how differences in bank fundamentals contribute to persistent regional disparities across the United States. Banks differ in their ability to screen borrowers, cost of capital, and demand for their loans, which in turn shape their ability to supply credit and on what terms. These differences therefore create a link between the geographic distribution of bank fundamentals and the geographic distribution of credit and economic activity. I will study this link, identify the mechanisms behind it, and examine how policy affects them, or how they may be used to as a policy tool.

A second project will study how the geographic overlap of banks' lending portfolios shapes the interaction between monetary policy and financial stability. Geographic overlap creates a two-sided relationship between the financial sector and local economies. First, banks operating across multiple regions may transmit shocks or any deterioration in their balance sheet across all its borrowers, creating a risk for the real economy if lending becomes concentrated among a small set of multi-regional banks. Second, these same banks can simultaneously be negatively impacted by an economic deterioration in a single region, presenting a concentration of risk in the financial system. I plan to use the bank-dynamics framework to study how this network of regional bank exposure affects the propagation of monetary policy and the evolution of financial stability over time.

Together, these projects extend my research to the interaction between bank dynamics, geography, and policy. My goal is to develop a unified framework for understanding how financial intermediaries allocate resources across firms and regions, and how that allocation shapes both short-run fluctuations and long-run economic outcomes.